Mahadi Hasan

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Research Interests

- **AI for Formal Verification:** using LLMs to synthesize verified code, proofs and invariants
- **Software Engineering & Programming Languages:** program correctness, verification tools

Education

University of Dhaka — Dhaka, BD
B.Sc. in Computer Science and Engineering — *July 2025*

- CGPA: **3.45/4.00**, top 30% of class
- Thesis Topic: Hypergraph Contrastive Learning for Financial Fraud Detection
- Supervisor: Prof. Chowdhury Farhan Ahmed
- Relevant Coursework: Theory of Computation, Discrete Mathematics, Software Engineering, Operating Systems

Research Experience

HIPREL Lab, University of Virginia — US, Remote
Research Intern — *Sept 2026 – Present*

- **Selected to continue** with **Prof. Wenxi Wang** after the UIUC++ SRSE program, one of only **9 of 107** students.
- **Proposed** a **first-author** project at the intersection of **Verus**, **LLMs** and **GPU kernels**, approved by Prof. Wang; now working toward a **paper submission**.

UIUC++ SRSE — US, Remote
Summer Research Intern — *June 2026 – Sept 2026*

- **Selected** for the UIUC++ SRSE 2026 program; studied researcher-filed bug reports with **Prof. Darko Marinov**, **acknowledged** in the ISSTA 2026 paper *The Discreet Charm of the Bugeoisie*.
- Built a pipeline that located and classified 258 bug reports behind an ISSTA 2025 paper's claims, matching them to CVE records and resolving 20 ambiguous cases by manual review.
- Contributed to **VeriContest**, a **NeurIPS 2026** dataset, by validating Verus specifications against natural-language descriptions and uncovering subtly flawed proofs; **acknowledged** in the paper.

Industry Experience

AfterQuery Experts — Contract · US, Remote
SWE, Testing & Eval — *Feb 2026 – Present*

- Contributed **Terminal-Bench** tasks in compilers & parsers for evaluating AI agents on real-world terminal workflows.
- Stress-tested task verifiers against **reward-hacking** exploits such as hard-coded outputs and reward forging, ensuring only correct solutions pass.

Samsung Research — Full-time · Dhaka, BD
Software Engineer — *Aug 2025 – Jan 2026*

- Developed **OpenCL** GPU kernels for IQ interpolation, notch filtering and gain control in a **B-mode ultrasound** signal processing pipeline.
- Built as tooling for a **Samsung Medison AI** research project on ultrasound imaging.

Turing — Contract · US, Remote
Research Analyst — *Sept 2024 – July 2025*

- Reviewed **Chain-of-Thought (CoT)** annotations, verifying each reasoning step so LLMs could learn **multi-step mathematical reasoning**.
- Audited other contributors' solutions and provided corrections on flawed reasoning.

Awards & Achievements

Math & Programming

- **National Math Olympiad:** Top 16 of more than 75,000 students, qualified for the IMO Team Selection Test — 2019
- **ACM ICPC Dhaka Preliminary:** Top 4%, qualified for the Dhaka Regional — 2021
- **Code Samurai Hackathon:** Finalist, one of 46 teams from 27 universities nationwide — 2024
- **Meta Hacker Cup:** Top 20% globally, qualified for Round 1 — 2020
- **Integration Bee:** Winner of the intra-university contest at the University of Dhaka — 2023

Academic Distinction

- **University of Dhaka Entrance Test:** Top 0.1% of more than 85,000 candidates — 2019

Teaching & Service

National Math Olympiad Dhaka, BD
Academic Team Member *Jan 2020 – Aug 2022*

- Graded contestants' answer scripts.
- Trained contestants at olympiad camps.

Battle of Brains, University of Dhaka Dhaka, BD
Co-host & Problem Setter *2023*

- Co-hosted the department's ICPC-style programming contest.
- Contributed an original problem, *Schrödinger's Brightness*, to the contest's problem set.

Udvash Dhaka, BD
Physics Instructor *Dec 2021 – Dec 2023*

- Taught physics classes to students preparing for top engineering university entrance tests.
- Mentored students on exam preparation and problem solving.

Selected Projects

BareMetal OS | *C, ARM Assembly, STM32F4* Coursework, 2023

- Built an operating system from scratch for an **ARM Cortex-M4** with no vendor HAL, C library or RTOS: boot code, interrupt and fault handling, and a **system call** interface separating user and kernel mode.
- Implemented **preemptive scheduling** through context switching between thread control blocks, and **counting semaphores** for synchronization.

EcoSync | *FastAPI, PostgreSQL, Next.js* Code Samurai, 2024

- Co-developed a waste management admin panel for Dhaka North City Corporation in the two-day **Code Samurai** final, with a **FastAPI** and PostgreSQL backend and a **Next.js** dashboard.
- Built JWT authentication and **role-based access control** across four user roles, with REST APIs for user, role and permission management, deployed with **Docker Compose**.

Skills

Programming Languages: C/C++, Python, C#, Rust, ARM Assembly

Verification & Systems: Verus, Z3, CUDA, OpenCL, ARM Cortex-M firmware

Frameworks & Tools: PyTorch, Git, Linux, Docker, LaTeX

References

Prof. Wenxi Wang wenxiw@virginia.edu
Assistant Professor of Computer Science, University of Virginia *PI, HIPREL Lab*

Prof. Chowdhury Farhan Ahmed farhan@du.ac.bd
Professor of Computer Science and Engineering, University of Dhaka *Undergraduate Thesis Supervisor*